BIOL 220 Graphics

Ka hua ʻōlelo o ka haʻawina

kuhikuhi

Learning Objectives

1. Distinguish between exploratory and explanatory figures
2. Identify what makes a good graph
3. Identify best practices in figure design
4. Identify misleading graphics and fix them
5. Understand how variable types determine figure design

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We make use of the *R* package `ggplot2` for figure design

The basic building blocks are:

```
library(ggplot2)

ggplot(data = "the data.frame we're plotting",
       mapping = aes(x = "column with data for x-axis",
                     y = "column with data for y-axis")) +
    geom_"type of plot"
```

5. Understand how variable types determine figure design

Explanatory:

Numerical

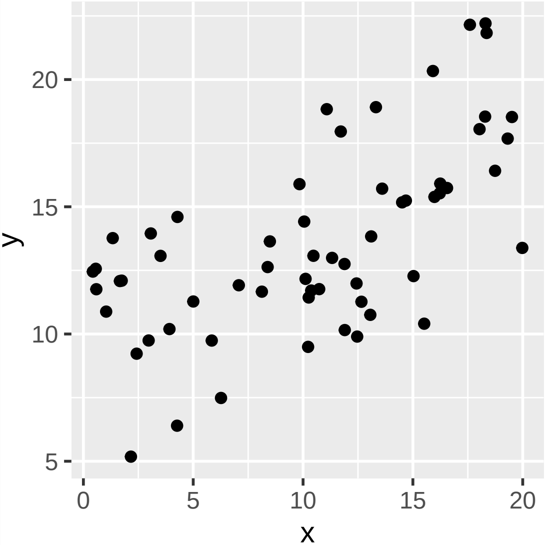
Response:

Numerical

Figure:

Scatter plot,

`geom_point`



5. Understand how variable types determine figure design

Explanatory:

Categorical

or Ordinal

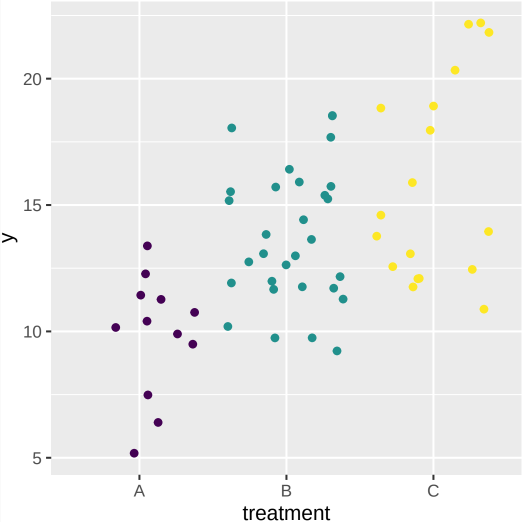
Response:

Numerical

Figure:

Jitter plot,

`geom_jitter`

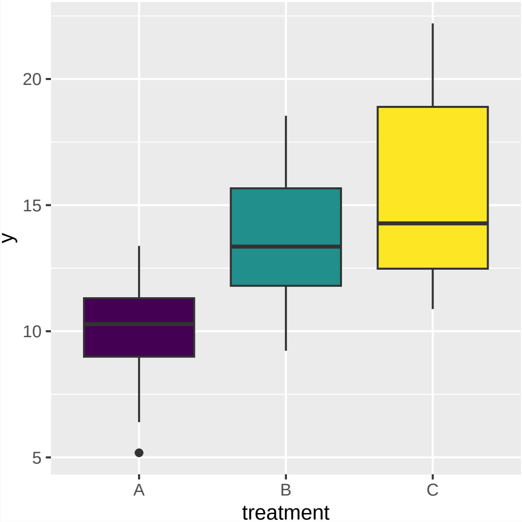


5. Understand how variable types determine figure design

Explanatory: Categorical *or* Ordinal

Response: Numerical

Figure: Box plot, `geom_boxplot`



5. Understand how variable types determine figure design

Explanatory:

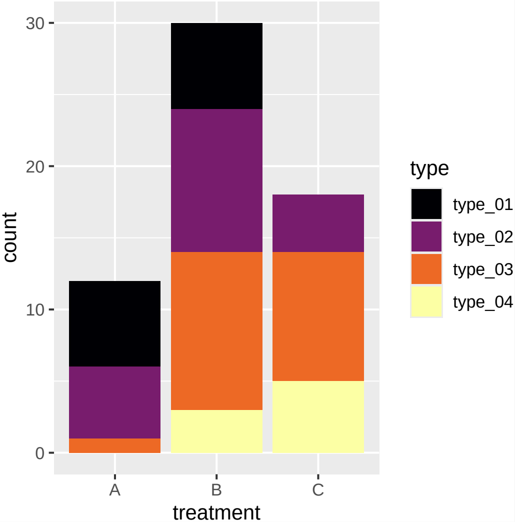
Categorical
or Ordinal

Response:

Categorical
or Ordinal

Figure:

Stacked bar plot,
`geom_bar`



5. Understand how variable types determine figure design

Explanatory:

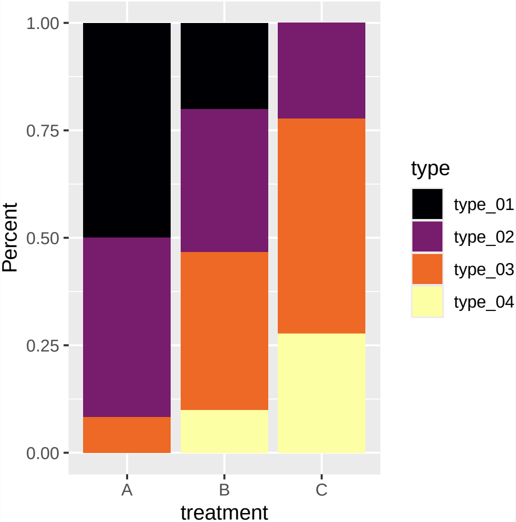
Categorical
or Ordinal

Response:

Categorical
or Ordinal

Figure:

Stacked bar plot (scaled to percent), `geom_bar(position = "fill")`



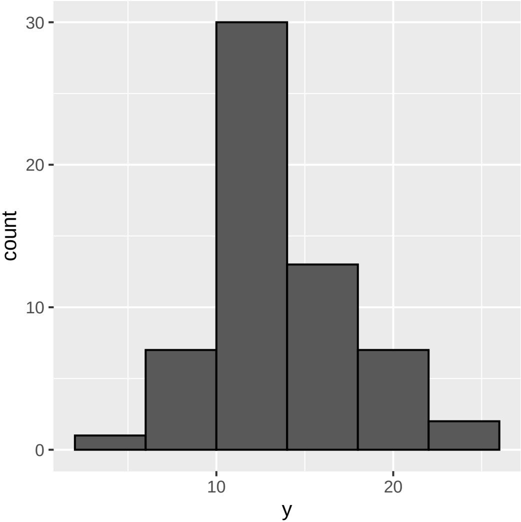
5. Understand how variable types determine figure design

Explanatory:

"none"
(i.e. you just want to check out response)

Response:
Numerical

Figure:
Histogram,
`geom_histogram`

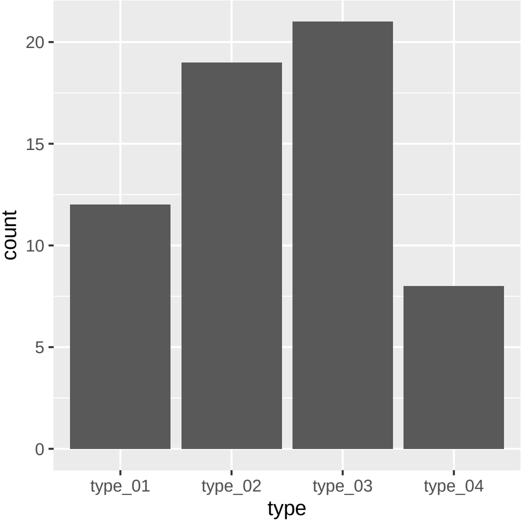


5. Understand how variable types determine figure design

Explanatory:
"none"
(i.e. you just want to check out response)

Response: Categorical *or* Ordinal

Figure: Bar plot, `geom_bar`



1. Distinguish between exploratory and explanatory figures

For exploratory plots: Don't worry about pretty colors, labels, names, etc. The goal is for you to understand the data.

For explanatory plots: Focus on all of this. You must consider your audience may, for example, consist of:

- People unfamiliar with your data
- People with different vision, and/or black-and-white printing

1. Distinguish between exploratory and explanatory figures

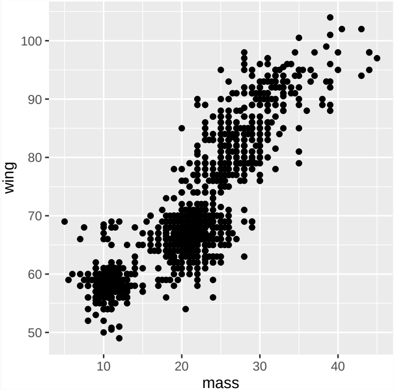
Let's use an example

```
# Data on morphology of some non-native birds in Hawai i
# from Gleditsch & Sperry (2020) *Ecology and Evolution*
birds <- read.csv("../../data/bird_morpho.csv")
head(birds)
```

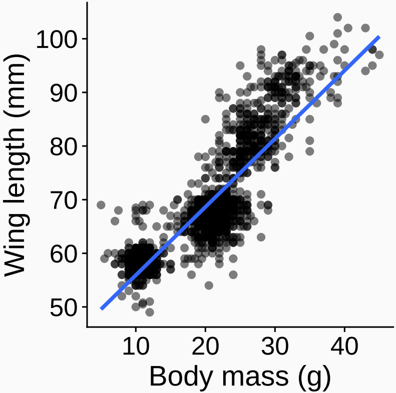
```
##   species              full_name sex tail tarsus wing nare
## 1    rble red-billed leiothrix   u   61   24.7   70
## 2    rble red-billed leiothrix   u   56   23.5   66
## 3    rwbu red-whiskered bulbul   u   81   20.6   80
## 4    rwbu red-whiskered bulbul   u   79   20.4   80
## 5    rwbu red-whiskered bulbul   u   77   17.5   76
## 6    rwbu red-whiskered bulbul   m   77   18.5   80
```

1. Distinguish between exploratory and explanatory figures

Exploratory



Explanatory



1. Distinguish between exploratory and explanatory figures

Contrast code using **ggplot2** (we fully explain in Lab 02)

Exploratory

```
# minimal code needed
ggplot(birds, aes(x = mass, y = wing)) +
    geom_point()
```

Explanatory

```
# plenty more code needed
ggplot(birds, aes(x = mass, y = wing)) +
    geom_point(size = 2, alpha = 0.5) +
    xlab("Body mass (g)") +
    ylab("Wing length (mm)") +
    geom_smooth(method = "lm", formula = y ~ x,
                se = FALSE) +
    theme_cowplot() +
    theme(axis.text = element_text(size = 18),
          axis.title = element_text(size = 20))
```

Keys to Good Data Visualization

Four keys of good data visualization

1. Show the data
2. Make patterns in the data easy to see
3. Represent magnitudes honestly
4. Display graphical elements clearly

Mistake 1: Hide the data

How to hide data:

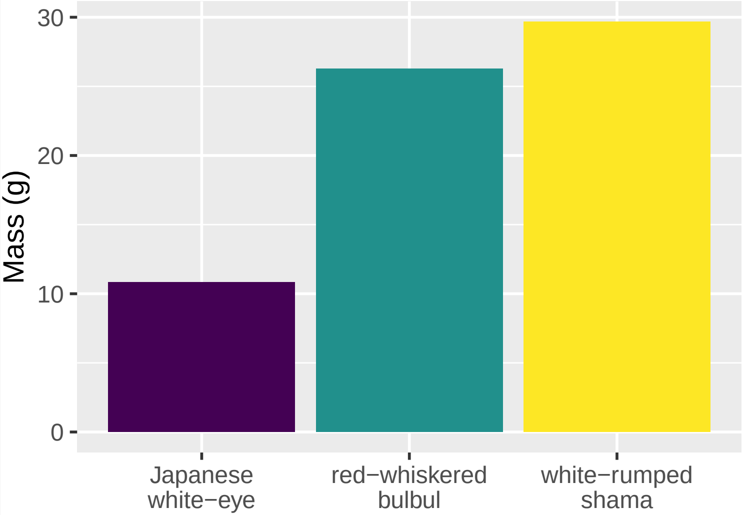
- Provide only statistical summaries
- Over-plotting

How to show data:

- Present all of the data points, while allowing all to be seen

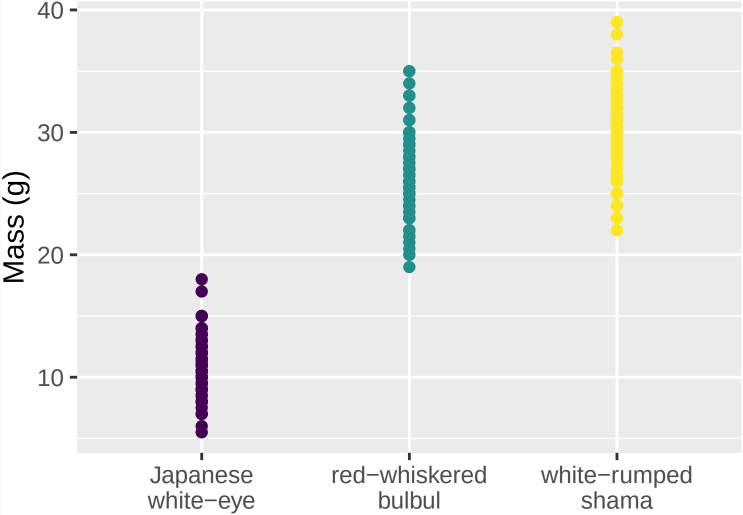
Mistake 1: Hide the data

This plot hides the variation within species



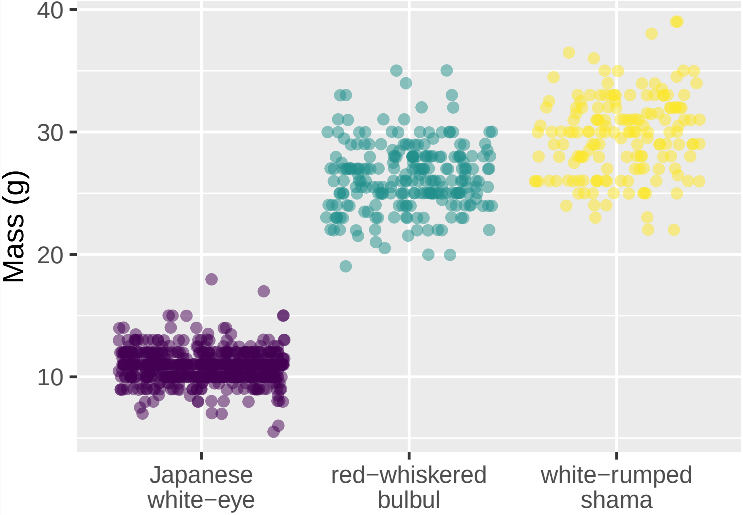
Mistake 1: Hide the data

This plot hides the density of observations



Mistake 1: Hide the data

Fixed! This plot shows all of the observations



Mistake 2: Make patterns hard to see

How to hide patterns:

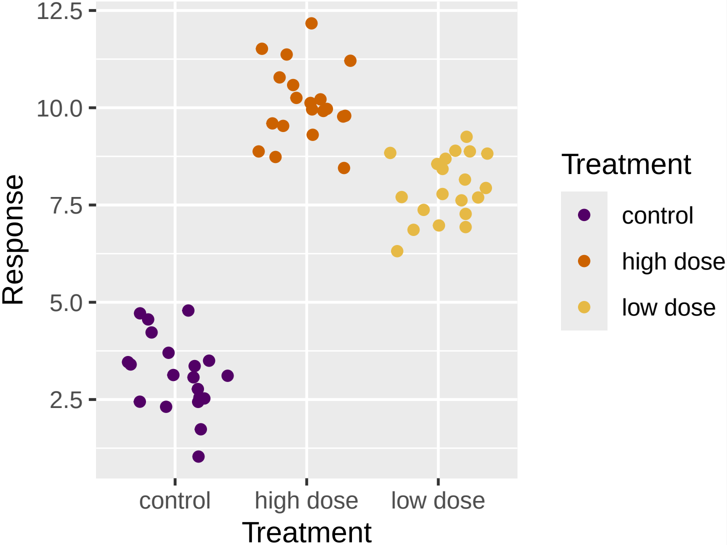
- Make one plot and call it good
- Use unreasonable scales
- Arrange factors nonsensically

How to make patterns in the data easy to see:

- Choose the best from multiple potential plots
- Use appropriate scales
- Arrange factors meaningfully

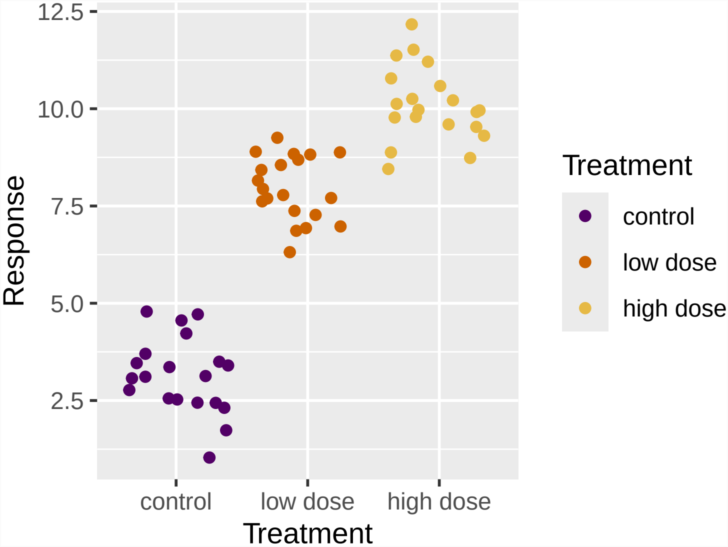
Mistake 2: Make patterns hard to see

Non-intuitive ordering of categories hides patterns



Mistake 2: Make patterns hard to see

Fixed! Intuitive ordering of categories emphasizes patterns



Mistake 2: Make patterns hard to see

Poorly chosen scale hides patterns

Mistake 3: Display Magnitudes Dishonestly

How to represent magnitudes *dis*honestly:

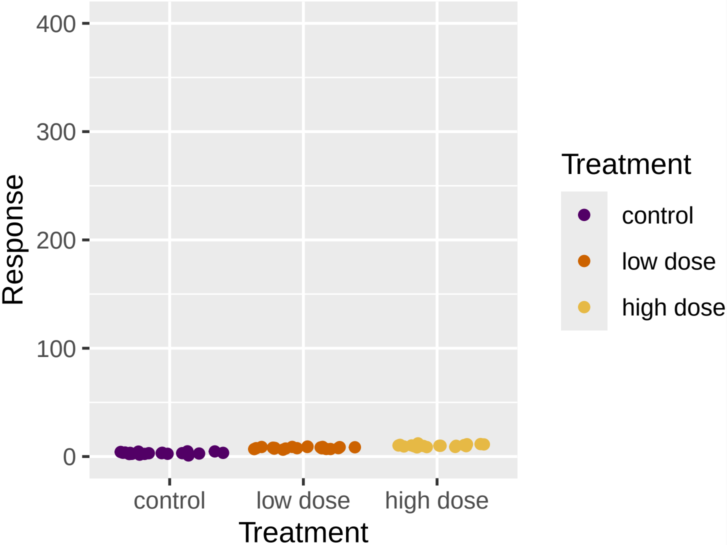
- Starting/stopping axes at arbitrary values

How to represent magnitudes *honestly*:

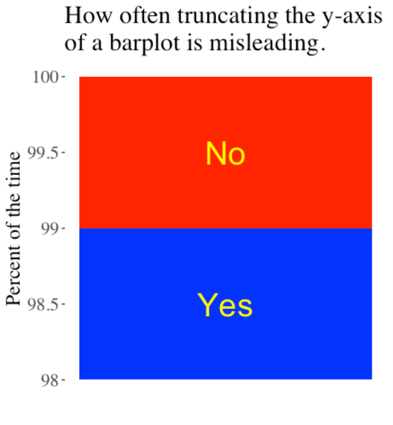
- Don't do that!

Mistake 3: Display Magnitudes Dishonestly

We've already seen this once



Mistake 3: Display Magnitudes Dishonestly



Note: there are a small number of cases where barplots can *honestly* start at a value other than zero

Mistake 4: Display graphical elements unclearly

How to display graphical elements unclearly:

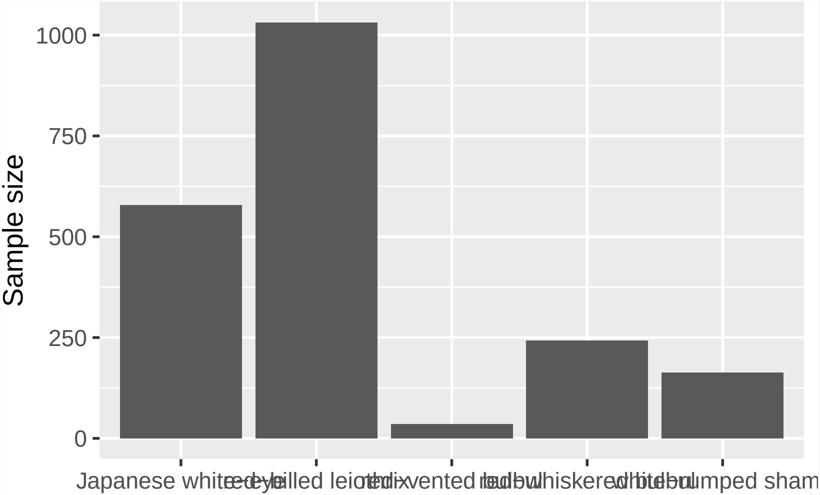
- Accept default plotting options
- Do not consider how a diverse audience will interpret your figure

How to display graphical elements clearly:

- Reflect on plot design
  - Are labels big enough?
  - Can all elements be clearly seen?
  - Is sizing and spacing appropriate?
- Consider diverse audiences and how they will interact with your plot

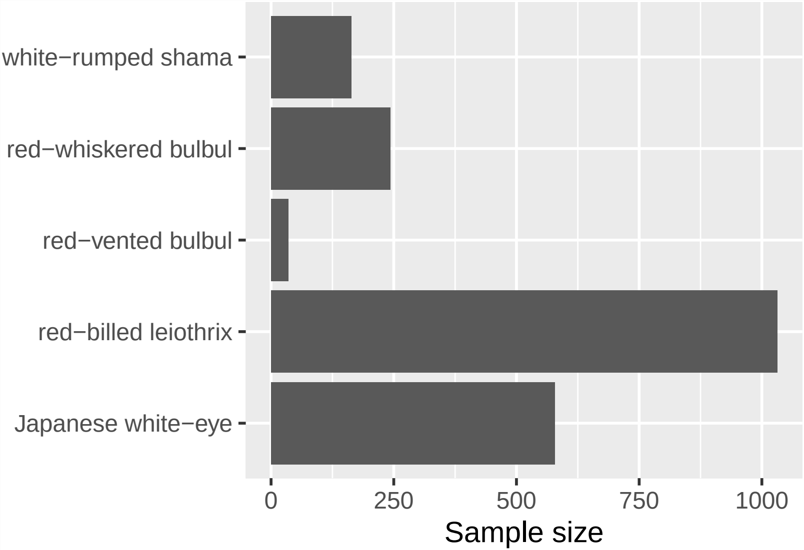
Mistake 4: Display graphical elements unclearly

Can you read the x-axis?



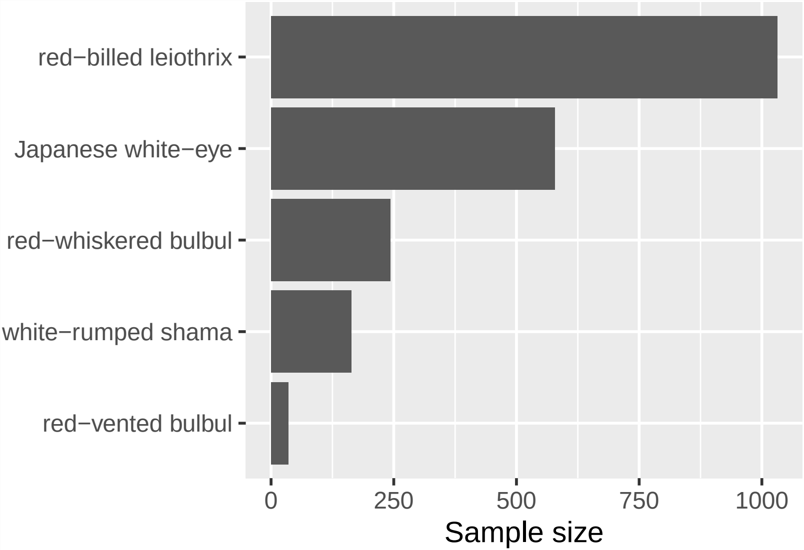
Mistake 4: Display graphical elements unclearly

Flipping improves readability



Mistake 4: Display graphical elements unclearly

Flipping improves readability...and we should probably re-order



Mistake 4: Display graphical elements unclearly

Consider colorblindness